

Do Returns to Schooling Differ by Race and Ethnicity?

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Inequality has increased in the United States over the past 25 years. In 1979, workers in the bottom 10th percentile of the wage distribution earned \$6.88 per hour (in 2003 dollars) while those at the 90th percentile earned \$30.19 per hour. By 1999 real wages at the 10th percentile had grown by roughly 5.4 percent to \$7.25. At the same time real hourly wages at the 90th percentile grew 18.1 percent to \$35.65. Because differences in human capital (education and experience) account for approximately one-third of the variation in wages and because the mean economic return to schooling is estimated at 10 percent, many researchers and policymakers have appealed to education and training policies to bolster the wages of the lowest-skilled workers (Pedro Carneiro and James J. Heckman, 2003; Alan Krueger, 2003).

Less is known about how the estimated returns to schooling vary across the population. For example, Carneiro et al. (2003) conclude that the return to education is higher for more-able individuals. In contrast, neither Joseph Altonji and Thomas Dunn (1996) nor Orley Ashenfelter and Rouse (2000) find consistent evidence that returns to schooling are higher for individuals that come from more-advantaged families. And yet, much social policy hinges on what we believe to be the value of education for subpopulations. For example, policies aimed at raising the incomes of the least-skilled members of society will neither improve their economic

well-being nor decrease inequality if their returns to schooling are low.

We provide further evidence on the variation in returns to schooling by examining whether the benefits vary by race and ethnicity of the individual. We do so by estimating returns to schooling using the U.S. Decennial Census as well as the National Longitudinal Survey of Youth, 1979 (NLSY79). We find that the return to schooling is relatively constant across racial and ethnic groups, even controlling for ability and measurement error biases.¹

I. Empirical Framework

Following Jacob Mincer (1974) we estimate the relationship between schooling and income by regressing the logarithm of the hourly wage ($\ln w_{ij}$) of individual i of race or ethnicity j on years of completed schooling (S_{ij}) controlling for explanatory variables such as potential experience or age, sex, race, and geographic region of the country (X_{ij}). As such we estimate

$$(1) \quad \ln w_{ij} = \alpha_j + \beta_j S_{ij} + X_{ij} \gamma_j + \varepsilon_{ij}$$

(where ε_{ij} is an error term). The coefficient on the schooling variable (β_j) is interpreted as the percentage increase in the hourly wage associated with one additional year of schooling and is referred to as the "return to schooling."

Related to many of the econometric issues raised below is the question of why we may or may not expect to find differences in the estimated return to schooling by race or ethnicity; that is, why β_j may vary by j . Begin by assuming that equation (1) represents the true relationship between wages and schooling and that differences in educational attainment occur randomly. Then, a constant β_j ($\beta_j = \beta$) implies that we should estimate the same return to schooling for any subgroup of the population. However, even if β_j is constant we may estimate different

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¹ See Barrow and Rouse (2005) for a more detailed discussion of this material.

returns to schooling for different subgroups if years of schooling is a poor proxy for human capital due to differences in school quality and if average school quality varies systematically by race. Alternatively, β_j may not be constant. For example, the return to schooling may depend on the level of education. In this case, estimating returns to schooling for subgroups of the population with different levels of education (on average) will generate different estimates of the return to schooling. For these reasons, it is an empirical question whether the return to schooling is constant across the population.

More generally, much of the literature estimating returns to schooling is concerned with whether the ordinary least squares (OLS) estimates of β_j in equation (1) reflect the causal effect of education on wages. Specifically, does more schooling cause higher earnings, or are more-able people more likely both to get more schooling and to earn higher wages, even in the absence of additional schooling? In the latter case, the OLS estimates of β_j will likely be upward biased due to selection on ability.

Further, selection bias may not be constant across the population. If African-Americans and Hispanics attend poorer-quality schools on average, then those students who manage to get more schooling may have unusually high ability. This selection would generate greater bias among African-Americans and Hispanics suggesting that the cross-sectional estimate of the return to schooling would be more upwardly biased for these populations. We address the potential for selection bias by including test scores as a direct control for ability and using family relationships by studying siblings.

Measurement error in reported schooling poses another econometric challenge. Attenuation bias caused by (classical) measurement error is exacerbated in within-sibling estimators, as identified by Zvi Griliches (1977), because sibling education levels are highly correlated. Thus, within-sibling estimates of the return to schooling are downward-biased. If the measurement error is classical in nature (i.e., uncorrelated with the error term in the wage equation and the true level of schooling) then an instrumental variables (IV) estimator using an independent report of the respondent's schooling as the instrumental variable will generate consistent estimates of β_j .

Further, measurement error may not be con-

TABLE 1—ESTIMATED RELIABILITY RATIOS FOR SCHOOLING LEVELS AND WITHIN-SIBLINGS, BY RACE AND ETHNICITY

Group	Levels	Within-sibling
All	0.89	0.74
African-American	0.81	0.67
Hispanic	0.86	0.77
Not African-American/ not Hispanic	0.92	0.78

Notes: Estimates are based on sibling data from the NLSY79. The within-sibling education is the deviation of the individual's schooling level from the mean education of his or her siblings.

stant across the population. The reliability (or "signal-to-noise") ratio is the proportion of the observed variance in schooling due to variance in "true" schooling. The correlation between two independent estimates of an individual's schooling level provides an estimate of the reliability ratio. Table 1 shows estimated reliability ratios for self-reported schooling, both in levels (i.e., for each individual) and for the deviation from sibling means. Overall we estimate that 11 percent of the observed variance in schooling levels is due to measurement error; however, our estimates vary by race/ethnicity. Nearly 20 percent of the observed variance in schooling for African-Americans is due to error, compared to 14 percent for Hispanics and 8 percent for whites. We also estimate more "noise" in sibling differences in educational attainment than in levels. Overall, 26 percent of the variance in sibling differences in education is due to measurement error, although the proportion due to error is one-third for African-Americans. Based on these estimates, we expect estimated returns to schooling for African-Americans and to some extent Hispanics to be more downward biased than for non-African-Americans/non-Hispanics.

II. Data

We begin using data from the 5-percent samples of the 1980, 1990, and 2000 Decennial Censuses. We limit the estimation sample to individuals aged 25–65 who were U.S. citizens, born in the United States, worked at least one week in the previous year, and earned at least one-half of the minimum wage. The regression analysis is based on annual earnings, and obser-

variations are weighted using Census individual weights.²

The NLSY79 is a survey of youth aged 14–21 in 1979. We use the 1993 wave in the analysis below and limit our sample to those with hourly pay greater than one-half of the minimum wage in 1993 and less than \$300 per hour, as well as those who are not self-employed, enrolled in school, or currently in the military. An advantage of the NLSY79 is that in 1980 most survey participants were administered the ASVAB, a basic skills test, from which it is possible to construct an Armed Forces Qualification Test (AFQT) score. While researchers disagree about whether AFQT scores mostly reflect “innate intelligence” or also reflect skills acquired in school, most would agree that they reflect *some* information about the skills that individuals possess at the time of the test. We use the AFQT as a measure of “observed” ability.³

Many NLSY79 respondents have siblings who are also included in the survey. In 1993, respondents were asked about their educational attainment as well as the educational attainment of up to 13 siblings. As a result, we have own-reported and sibling-reported measures of a respondent's education level for those with siblings in the original NLSY79 sample who also participated in the 1993 wave. Excluding those with no wage and education information, our sibling sample contains data on 2,419 individuals from 1,062 households.

III. Results

Using Census data we generate very precise estimates of the relationship between education and wages by race/ethnicity. In addition, we

² We get greater variation when we estimate the returns to schooling using hourly wages rather than annual earnings. This is because the relationship between greater schooling and more stable jobs is stronger for African-Americans and Hispanics than for nonminorities. Whether this correlation is explained by access to more stable jobs or changes in labor supply decisions is an empirical question, but one we cannot address here.

³ In the estimates presented here, we simply control for AFQT and do not address the fact that individuals took the AFQT at different ages and had therefore completed differing years of schooling. We have also estimated our models controlling for the individual's education as of 1979 with qualitatively similar results.

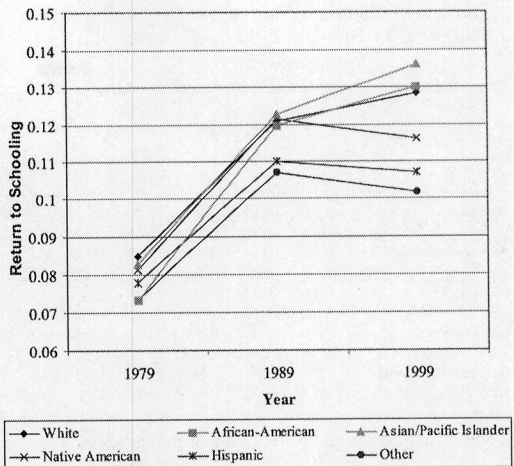


FIGURE 1. ESTIMATES OF THE RETURNS TO SCHOOLING BY RACE AND ETHNICITY USING U.S. DECENNIAL CENSUS DATA

Notes: Estimates use U.S. Decennial Census Data from 1980, 1990, and 2000 and are based on a regression of the logarithm of annual earnings on years of schooling, indicators for nine regions, a quadratic in potential experience, and an indicator for whether the individual was female. Observations are weighted using Census individual weights.

look at how the relationships have changed during a period of large increases in both inequality and the returns to schooling. The primary drawback of Census data is that we cannot examine the potential for ability bias or measurement error problems.

In Figure 1 we present estimates of the returns to schooling by six race/ethnicity categories.⁴ In 1979, an additional year of schooling is associated with a 7.3-percent increase in annual income for African-Americans and an 8.5-percent increase in annual income for whites. Between 1979 and 1989 the estimated return to schooling increased dramatically for all races, especially African-Americans, but remained in a relatively tight range: 10.7–12.3 percent. Between 1989 and 1999 the increase in the estimated returns to education is much smaller on average, but the range of estimates is larger (10.2–13.6 percent).

⁴ These returns to schooling were estimated from OLS regressions of the logarithm of annual earnings on years of schooling, indicators for nine regions, a quadratic in potential experience, and an indicator for whether the individual was female. The regressions were weighted by the Census weight.

TABLE 2—OLS AND IV ESTIMATES OF THE RETURNS TO SCHOOLING ($\times 100$) USING SIBLINGS IN THE NLSY 1979 COHORT

	Cross section		Within siblings			
	OLS		OLS		IV	
	(i)	(ii)	(iii)	(iv)	(v)	(vi)
Overall	9.36 (0.40)	6.15 (0.48)	7.60 (0.69)	5.21 (0.76)	9.16 (0.88)	6.74 (1.02)
African-American	9.97 (0.73)	7.14 (0.89)	9.84 (1.23)	6.60 (1.38)	12.66 (1.71)	9.87 (2.12)
Hispanic	9.18 (1.05)	5.92 (1.21)	7.21 (1.60)	5.32 (1.76)	8.52 (1.91)	6.45 (2.20)
Not African-American/ not Hispanic	9.17 (0.54)	5.64 (0.66)	6.30 (1.00)	4.44 (1.07)	7.54 (1.24)	5.65 (1.38)
Include AFQT score	N	Y	N	Y	N	Y
Sibling fixed effect	N	N	Y	Y	Y	Y

Note: Standard errors are reported in parentheses.

Based on these estimates, we would conclude that the estimated return to schooling for African-Americans is roughly the same as that for whites and that the return for Hispanics is somewhat lower.

We present estimates of the return to schooling using the NLSY79 in Table 2. Each cell represents a separate estimate of the return to schooling ($\times 100$). The basic specification is an OLS regression of the natural logarithm of hourly pay on years of completed education, a third-order polynomial in age, an indicator for female, indicators for four geographic regions, and a constant. These regressions are unweighted, although results are similar if we weight. Further, we do not cluster the standard errors on the household such that the standard errors, especially those within sibling, are understated.⁵

Each column represents a different specification using the sibling sample. The specifications in columns (i) and (ii) are cross-sectional. The estimates in the odd-numbered columns do not include the AFQT score while those in the even-numbered columns do. The estimates in columns (iii)–(vi) control for a sibling fixed effect, and those in columns (v) and (vi) use the average of the sibling-reports of the respondent's education as an instrumental variable in an IV analysis.

⁵ In the cross-sectional specifications, the standard errors using "cluster" are very similar to those that do not allow for such intra-household correlations; the standard errors for the within-sibling specifications presented in columns (iii) and (iv) in Table 2 are understated by about 50 percent.

Our overall estimate of the cross-sectional return to schooling is about 9 percent. The estimate is highest among African-Americans (10 percent). In general, controlling for selection by including AFQT scores decreases the cross-sectional estimate of the return to schooling by about 3 percentage points [comparing columns (i) and (ii)]. The effect on estimates for Hispanics and whites is somewhat larger than for African-Americans. Based on these cross-sectional estimates, we find little difference in the return to schooling by race/ethnicity.

Estimates that account for a sibling fixed effect are presented in columns (iii)–(iv). A comparison of the estimates in columns (i) and (iii) [or columns (ii) and (iv)] suggests that controlling for a sibling fixed effect makes a bigger difference for nonminorities than for minorities, especially African-Americans. In fact, the within-sibling estimate of the return to schooling for African-Americans is at most 0.5 percentage points lower than the cross-sectional estimate. The fact that controlling for siblings makes a smaller difference for minorities than nonminorities may reflect less selection bias in the cross-sectional returns to schooling. Or, it may suggest that controlling for a sibling fixed effect is less effective for some populations than others. Why might this occur? While we limit our sample to "siblings" (excluding foster siblings, step-siblings, and adopted siblings), we cannot distinguish between "full siblings" and "half-siblings." If African-Americans and Hispanics are more likely to live with half-siblings than are nonminorities, then the family fixed effect may not be a good proxy for unobserved family "ability."⁶

Finally, the overall measurement-error-corrected (IV) estimate of the within-sibling return to schooling increases to 9.2 percent in column (v) relative to an estimate of 7.6 percent in column (iii) suggesting an attenuation bias of about 20 percent. Further, we find that correcting for measurement error has the greatest effect

⁶ As evidence consistent with this explanation, the within-sibling coefficients in column (iv) (that control for AFQT) decrease the most relative to column (iii) (that do not control for AFQT) for African-Americans. This pattern of results would be expected if the AFQT controls for other aspects of "ability" that are not captured by the sibling fixed effect.

on the estimated returns to schooling for African-Americans, as expected based on the reliability ratios in Table 1. Generally, while we continue to estimate a larger point estimate for African-Americans than for Hispanics and nonminorities, the differences across race/ethnicity are not statistically significant.⁷

IV. Conclusion

Using data from the Census and the NLSY79, we find little evidence of differences in the return to schooling across racial and ethnic groups, even with attempts to control for ability and measurement-error biases. While we find point estimates that are relatively similar across racial and ethnic groups, our conclusions are partly driven by large standard errors in some specifications due to small sample sizes. That said, we find no evidence that returns to schooling are lower for African-Americans or Hispanics than for nonminorities. As a result, policies that increase education among the low-skilled, who are disproportionately African-American and Hispanic, have a good possibility of increasing economic well-being and reducing inequality.

We find some evidence that measurement error and selection bias may differ by race and ethnicity. For example, self-reported levels of schooling are "noisier" for African-Americans than for other groups. And, we find less evidence of ability bias among African-Americans and Hispanics than among nonminorities. More generally, our analysis suggests further research is needed to better understand the nature of

measurement error and ability bias across subgroups in order to fully understand potential heterogeneity in the return to schooling across the population.

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⁷ In results not presented here, we have also estimated these specifications using annual earnings rather than hourly wages. While the estimated returns to schooling are a little higher, there are no differences by race or ethnicity.